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Supply chain risk management for critical commodities: A system dynamics model for the case of the rare earth elements

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ABSTRACT

We investigate mitigation strategies that potentially reduce the supply chain risk caused by exogenous interventions for critical commodities. We apply a system dynamics approach to examine the individual company's reaction in the rare earth elements (REEs) supply chain to China's introduction of export restrictions over a three-stage supply chain for 11 REEs, 10 semi-finished goods (SFG), and 21 finished goods (FG) between 2004 and 2015. We analyze how FG and SFG substitution, secondary recycling, supply extension and research and development (R & D) affect dynamic variables such as the price level, supply and demand on each stage to determine the effectiveness of each strategy for mining companies, SFG and FG producers.

FG and SFG substitution show the highest potential for mitigating supply unavailability. Substitution is especially beneficial for FG producers. It also has the advantage of offering a quick development of alternatives for reducing REE supply unavailability risk. Due to the fact that the necessary infrastructure and technology are not available, secondary recycling shows only a low mitigation potential for FG companies. It was shown that mining companies do not invest proactively in supply capacities beyond China, as our simulation shows that it took 6 years until a new REE mine was opened. R & D is identified as a key element for REE mining companies, SFG and FG producers when it comes to developing alternative products and recycling techniques. Nonetheless, SFG producers face huge difficulties when trying to develop substitutes as alternative materials are not present.

1. Introduction

Manufacturers world-wide experienced a serious and abrupt supply shortage of rare earth elements (REE) in 2006. China, the major supplier of REE with a market share of 98% at the time, introduced export restrictions that limited global supply dramatically (Reichl et al., 2015; Massari and Ruberti, 2013). The supply drop triggered an exogenous intervention for all downstream companies operating in the REE supply chain. Due to the market power of China, no alternative supplier was available. Furthermore, REEs have very specific characteristics that make a substitution of the commodity nearly impossible unless firms are willing to accept a massive loss (Schüler et al., 2011).

Besides REEs, there are other commodities for which limited supply and high risk are also a problem. The US National Research Council (2008) introduced a two-dimensional "criticality matrix" covering the "importance in use" and "availability" of commodities. They analyzed and classified "critical" 20 raw materials out of a list of 54 candidates for the US economy, for example Cobalt, Graphite, Magnesium, Platinum Group Metals, REE, Tantalum or Tungsten. Further analyses that give REE the highest supply risk of all critical commodities (European Commission, 2014; United States Geological Survey, 2014; Moss et al., 2013; Behrens et al., 2007) have been carried out. Nassar et al. (2012) enhance the "criticality matrix" by adding a third dimension: "environmental implications". The "critical space" shows results similar to those of the "criticality matrix" (Nassar et al., 2015).

In situations such as the REE intervention, managers are forced to mitigate the supply chain risk of the company while at the same time having to meet the demand from customers. This typically leads to strategic decisions since, due to the market power of China, the options are limited in the case of REEs. Substitution (reducing REEs, semi-finished goods or finished goods demand), recycling (recollecting REEs from existing finished goods), the opening of new mines (increasing REE supply) or research and development (R & D) activities (developing alternative products and materials, recycling techniques) are some of the potential solutions. Nevertheless, these strategic options have operational consequences: the development of substitutes and the redesign of current products, the setup of recycling infrastructures, changes in current R & D plans and workforce, along with manufacturing processes are decisions that require significant time and investments. If export restrictions are only temporary and supply will again be

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available in the future, unnecessary losses of time and cash, as well as operational and organizational disturbances, will occur. Therefore, knowledge on how the underlying dynamics of REE supply impacts the entire REE supply chain is needed.

The analysis of these underlying effects is part of the systems thinking theory developed by Forrester (1958). The systems thinking approach covers different stakeholders and shows how they interact. System dynamics (SD) is based on system thinking and allows the analysis of the cause-and-effect relationship between decisive factors such as the price level reaction and the change of the potential demand at each stage of the model. Feedback loops that develop each variable individually can be established (Forrester, 1958). SD reflects the driving factors as well as the sub-systems in a standardized structure and analyzes their interactions, behavior and interdependencies (Senge and Sterman, 1992).

SD has already been applied to several areas and industries and allows insights, also with respect to socio-economic systems (Besiou et al., 2011; Senge and Sterman, 1992; Morecroft, 1988; Forrester, 1971), the automotive industry (Walther et al., 2010), energy and environmental markets (Ford 2010; Morecroft and van der Heijden, 1992), (financial) supply chain systems (Udenio et al., 2014; Geman and Ohana, 2008), supply chain risk management (Ghadge et al., 2013; Zamora Aguas et al., 2013; Zsidisin and Ritchie, 2008) and commodity markets (Asif et al., 2015; Sverdrup et al., 2014a; Kifle et al., 2013; Rogetzer, 2012; Greenman, 1994). SD can represent market behavior and show insights into the market mechanism.

With regard to REEs, Kifle et al. (2013) analyzed how long REE supply will last (in the next five thousand years) and whether there is any risk of running out of an REE. Their results show that, in the next 30–300 years, suppliers will have to accept considerable risks if they wish to meet future supply. They suggest that the government could stabilize the market by requiring recycling activities from all companies. Rogetzer (2012) investigated how REE companies react if price volatility is high. She developed an REE model that simulates a supply-demand-ratio for analyzing scarcity. She shows in a multivariate sensitivity analysis that REE demand is inelastic even if commodity prices fluctuate strongly.

2. Methodology

2.1. Scope and boundaries

We focus on the development of an end-to-end supply chain that covers the underlying dynamic effects of 11 REEs, 10 semi-finished goods (SFGs) and 21 finished goods (FGs). SFGs are products that are based on REEs. The REEs are modified and transformed into a product, e.g. a magnet, which is then processed in the REE supply chain. FGs are based on already assembled SFGs. FGs are manufactured for the specific needs of the requesting industry, for instance a wind turbine for the renewable energy industry.

The introduction of export restrictions caused an exogenous shock through the entire REE supply chain. Downstream companies like SFG and FG manufacturers had to react and take strategic decisions. The REE model presented below examines five different mitigation strategies: (1) substitution, (2) recycling, (3) the opening of new mines, (4) R & D (the development of alternative products, processes, or new recycling techniques) and (5) the interplay of strategies (1) to (4). Strategy (1), (2) and (4) target individual companies on all three stages (REE mining companies, SFG and FG producers) of the REE supply chain. Strategy (3) focuses on REE mining companies and the overall REE industry from a macroeconomic point of view; it also examines how governments could reduce REE supply dependencies by supporting the opening of new REE mines.

The aim of this study is to analyze the impact of proactive (substitution, the opening of new mines and R & D) and reactive (recycling) strategies for avoiding or recovering from supply unavailability. We also want to analyze how the different mitigation strategies reduce the price level and the demand on the various stages of the REE supply chain. We are interested in identifying which mitigation strategy performs best in terms of lowering the absolute REE price after the exogenous intervention occurred (Zhang et al., 2015). Our research focuses on three specific questions:

1. What underlying dynamic effects are triggered within the REE supply chain by a substitution, a recycling, and a supply expansion strategy for REE mining companies, SFG and FG producers?
2. What effect does R & D have on REE, SFG and FG substitution for the respective companies and on secondary recycling for FG producers?
3. Which mitigation strategy (or combination of strategies) should mining companies, SFG and FG companies apply at their respective supply chain stage under what circumstances to mitigate the risk of REE supply unavailability?

Our work differs from previous studies in that it uses extensive empirical data for model parameterization and calibration and because it models an end-to-end three stage supply chain for the special case of REEs. As pointed out by the Bavarian Industry Association, covering the REE supply chain end-to-end is necessary to provide sustainable suggestions, since there is a high market concentration in the downstream manufacturing stages (Vereinigung der Bayerischen Wirtschaft, 2015).

The end-to-end supply chain model includes local, geographical and time restrictions. The local and geographical boundaries of this study are related to the stakeholders and countries we examined. REE mining companies, raw material processors, SFG and FG manufacturers, primary and secondary recycling firms, R & D labs, as well as 10 countries and consumers from 18 different industries (e.g. oil, automotive, ceramics), are included. Mining companies and raw material processors are modeled per country, whereas SFG and FG manufacturers, primary and secondary recycling firms and R & D labs are modeled on the supply chain level. Furthermore, political and economic interests are modeled on an aggregated level for a particular country or on a supply chain level, rather than on a single company/decision maker level (Udenio et al., 2014). All countries that mined REEs during the analyzed time period are included. The countries are Australia, Brazil, China, India, North Korea, Malaysia, Russia, South Africa, Sri Lanka and the United States of America. The consumers' demand from the 18 different industries is influenced by the growth rate of the requesting customer industry and the global population. Details are available in Appendix A and in the Supplementary Material. Our period of interest is the time between 2004 and 2015.

2.2. Key assumptions

The following assumptions were made for simplifying and data availability reasons:

- The model considers only REEs as a critical input material in the production of SFGs. To remain focused on REE, other raw materials are not explicitly modeled.
- The customer industries for each FG do not change over the simulation period. Changing demand is modeled through the possibility of substitution and increases due to the population growth (Population Reference Bureau, 2015; United Nations, 1998).
- The coefficients at which SFGs and FGs are produced do not change over the simulation period. Where no data is available and several FGs are based on one SFG, we assume an equal allocation.
- In their REE lifecycle analysis, Du and Graedel (2011) show that 10% of all REEs are lost during the separation process and another 10% during the fabrication of SFGs; we use the sum of the two values as the SFG scrap ratio.
- Smelting and concentration loss of REE oxide is 4%, based on the reduced mining output amount (China's Ministry of Industry and

Information Technology, 2015).

- The proportion of high-quality raw earth elements (HQ REEs) within the secondary recycling is 70% and 30% for low-quality raw earth elements (LQ REEs), respectively. LQ REEs have to be concentrated and smelted again, which is an assumption deduced from the global copper model (Glöser et al., 2013).
- During the semi-finished goods production, 1% of the raw material is unrecoverable (Glöser et al., 2013).
- Schüler et al. (2011) assumes that the primary recycling rate is 33%.
- Goonan (2011) states that it was possible to start secondary recycling in low quantities as early as 2008. We define low quantities with an initial rate of 5%. The fact that recycling was started in 2008 indicates that the life time of FGs is 48 months. Schüler et al. (2011) estimated that some FGs like electronic motors can have a life time of 5–10 years. We use 48 months for all FGs in our analysis.
- Secondary recycling activities will only be performed if a recycling technology is available. No new technology for other FGs is developed during the simulation period.
- The hiring and laying off process of the R & D workers is delayed by 6 months. The hiring duration is 23 working days on average (Chamberlain, 2015). Zachert (2004) shows that the notice time for making employees redundant depends on country-specific labor laws, on how long the worker has been employed by the company and on company-specific regulations. It varies between 1–24 months.
- The introduction of new basic research programs from politics is delayed by 12 months. The value is an average based on Leszczensky et al. (2004).
- The rate at which the market share of emerging and developed markets changes over time is linear over the simulated period.
- We assume an economic interest with a rate of 10%, as (1) the value could not be quantified, (2) a change over a 12-year time horizon is likely and (3) global company operation could bias the value.

2.3. System dynamics model and its applications

The three stage REE model contains 4 modules; (1) supply and demand, (2) R & D, (3) substitution and (4) primary and secondary recycling. The model is implemented in AnyLogic 7 and takes into account structural, operational, and behavioral parameters on a monthly basis at a single raw material level. We model a total of 8 supply chains, which are made up from 11 REEs, 10 SFGs and 21 FGs for 18 consuming industries from emerging and developed markets. Within each supply chain, several SFGs and FGs can be present. The three stages are REE production, processing and usage. Fig. 1 presents the causal loop diagram of one supply chain for the three stage system dynamics REE model. The explanation of the three stages and of how the supply chains interact can be found in Appendix A. Details regarding the supply chain allocation can be found in the Supplementary Material.

The supply and demand module exists in all three stages of the REE model. REE supply is based on the mined REE ore, which is then smelted and refined so that it is of high quality and can be used for downstream production. Available REEs are transferred to the SFG production. During these production processes, cutting and stamping activities are performed, which results in REE scrap. After the production of SFGs, the products are transferred to the FG production. Produced FGs are moved to the FG stock, where they are consumed by the customer industries. After the life time of the FGs, 5 FGs are available as FG scrap for secondary recycling activities. All other FGs are landfilled and lost.

The equations are organized as follows: (1) to (5) are related to generating supply and goods production. Absolute and potential demand is modeled in (6) to (9). (10) and (11) are related to the industrial and population growth and (12) to the price level calculation. Political, the companies' and economic interest is modeled in (13) and (14). In (15) to (17), it is illustrated how R & D know-how can be generated and the calculation of the substitution amount happens in (18). Finally, the calculations for primary and secondary recycling are presented in (19) to (21). An index of all the variables we used can be found in the Supplementary Material.

The total supply of each raw earth element includes the mined REE ore and recovered material from primary and secondary recycling (China's Ministry of Industry and Information Technology, 2015; Reichl et al., 2015; Christmann, 2014; United States Geological Survey, 2014; Glöser et al., 2013; Goonan, 2011). The REE ore is smelted and concentrated to a level that makes it possible to use the raw material for SFG production as detailed in (1):

$$\mathrm{REE\ S_{it}} = (1 - \mathrm{REE\ Loss}) \cdot (\mathrm{M_{it}} + \mathrm{REE\,SR_{it}} \cdot \mathrm{LQ\ REE\ SR}) - \mathrm{Gov\,R_{it}} + \mathrm{REE\ PR_{it}} + \mathrm{REE\ SR_{it}} \cdot \mathrm{HQ\,REE\ SR} \cdot (1 + \mathrm{P_{it}}) \quad (1)$$

where i are the different types of raw earth elements, semi-finished and finished goods used within the REE model; t is the time period of the REE model; $(1 - REE\ Loss) \cdot (M_{it} + REE\ SR_{it} \cdot LQ\ REE\ SR) - Gov R_{it}$ is the mined REE ore in tons and the recovered low quality REEs that have to be smelted and concentrated again, as well as the amount that is restricted due to a governmental statute; $REEP\ R_{it} + REE\ SR_{it} \cdot HQ\ REE\ SR$ are the recovered REEs out of primary and secondary recycling that are of high quality and can be reused directly; and $(1 + P_{it})$ is the current price level of the REE.

Supply for SFGs and FGs is based on the pre-defined bill of material coefficient or SFG coefficient necessary for manufacturing the good on the relevant stage. The specific coefficients are listed in the Supplementary Material. The supply of SFGs and FGs is expressed as follows:

$$\mathrm{SFG\ S_{it}} = \mathrm{REE\ F_i} \cdot \mathrm{REE\ S_{it}} \quad (2)$$

$$\mathrm{FG\ S_{it}} = \mathrm{SFG\ F_i} \cdot \mathrm{SFG\ S_{it}} \quad (3)$$

$REEF_i$ is the coefficient of the used REEs, $REES_i$ is the supply amount in tons of the specific raw material used for calculating the SFGs supply. Equation (3) presents the supply for FGs where $SFG\ F_i$ is the coefficient of the used SFGs and $SFG\ S_{it}$ is the supply amount in tons of the specific SFGs used for calculating the FGs.

During the production of SFGs, some of the supplied REEs are lost due to cutting and stamping activities. Part of the loss can be collected and recycled. The recollected REEs are fed back into the REE supply as primary recycled material. The production of SFGs is expressed as:

$$\mathrm{SFG\ Pro_{it}} = (\mathrm{SFG\ S_{it}} - \mathrm{SFG\ Scrap_{it}} - \mathrm{SFG\ S_{it}} \cdot \mathrm{SFG\ Loss}) \cdot (1 + \mathrm{P_{it}}) \quad (4)$$

$SFG\ S_{it} \cdot SFG\ Loss$ is the loss that is unrecoverable from SFGs supply and $SFG\ Scrap_{it}$ is the material available for primary recycling activities. During the production process of FGs, no losses occur. The FG production is expressed by the following equation:

$$\mathrm{FG\ Pro_{it}} = \mathrm{FG\ S_{it}} \cdot (1 + \mathrm{P_{it}}) \quad (5)$$

$FG\ S_{it}$ is the FGs supply.

There is a difference between potential demand and absolute demand at each stage. Absolute demand is based on the potential demand but reduced by substitutes. The absolute demand is explained by the following equation:

$$\mathrm{D_{it}} = \mathrm{PD_{it}} - \mathrm{Sub_{it}} \quad (6)$$

PD_{it} is the potential demand for the different types of REE, SFG and FG in tons and Sub_{it} the substitution amount in tons for the specific REE, SFG and FG. Note that a substitution of REEs is only theoretically possible, as was pointed out in the introduction.

Potential demand is first calculated at the FG stage based on empirical data from 2008 (Goonan, 2011). As a first step, the potential FG demand is defined based on population growth, industrial demand, and the current FG price level. The potential FG demand is then the input variable for the potential SFG demand, which is also influenced by the SFG price level. The potential SFG demand is the input variable for the

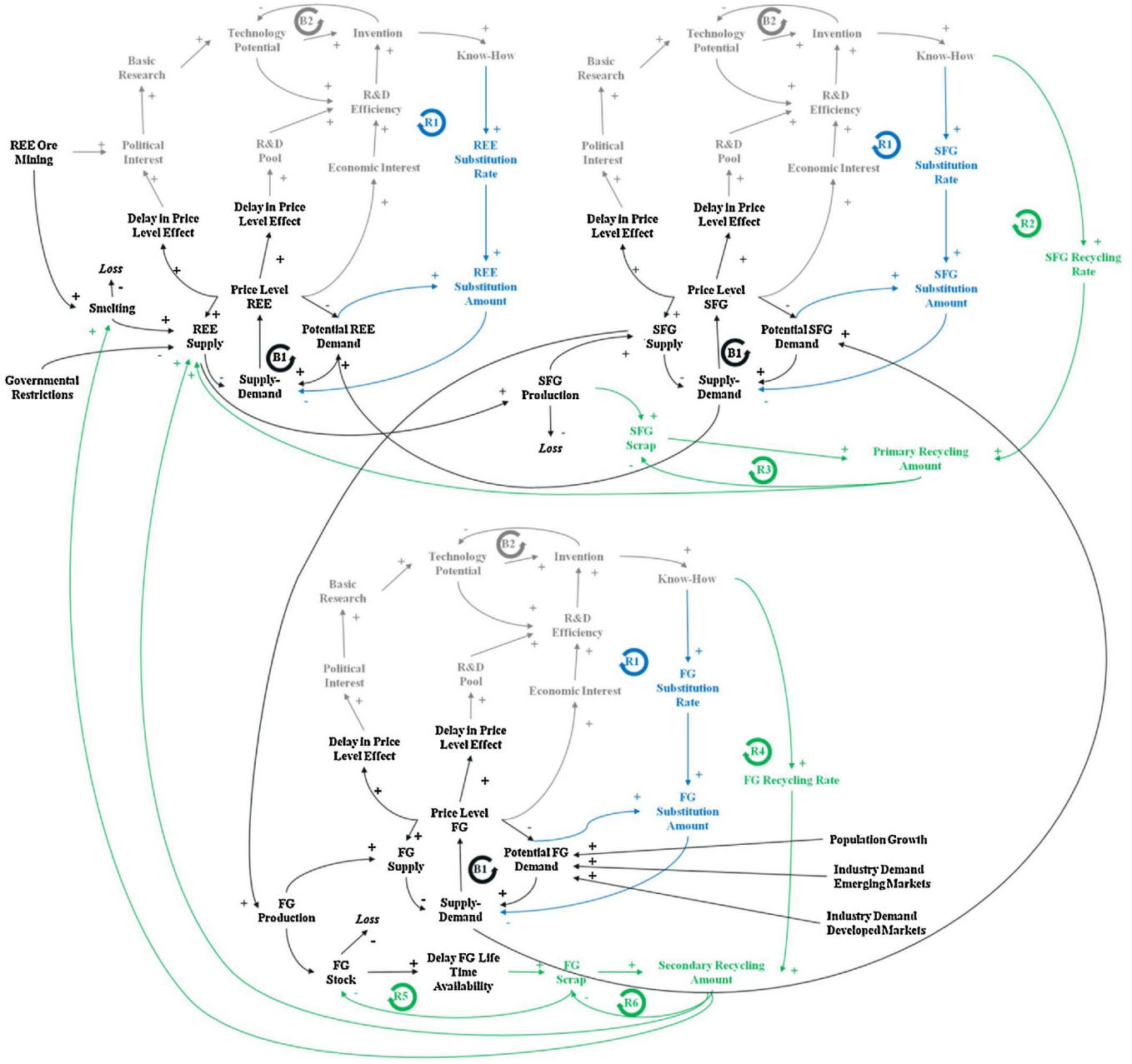


Fig. 1. Three stage system dynamics REE model (supply and demand are represented in black, R & D in grey, primary and secondary recycling in green and substitution in blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

potential REE demand that follows the price level calculation of the downstream stages. The approach for the SFG and REE calculations is similar. Potential demand is influenced at each stage of the REE model by the current price level of a product. The lower the supply of a good, the less absolute demand of the good can be fulfilled and hence, due to a shortage, the higher is the price level. A higher price level causes the potential demand to drop and supply to increase and vice versa.

The population growth is based on data from the year 2004. The industrial growth is weighted via emerging and developed markets to consider market dynamics, behavioral changes like the switch to renewable energies, support of electromobility or technological developments like wind turbines, glass altering and glass coloring and changes over the simulation period. The potential demand for FGs, SFGs and REEs is expressed by equations (7), (8) and (9):

$$\text{FG PD}_{it} = \text{D}_{i,t48} \cdot \text{IG}_{it} \cdot \text{PG}_{t} \cdot (1 - \text{P}_{it}) \quad (7)$$

$$\text{SFG PD}_{it} = \text{FG D}_{it} \cdot \text{SFG Fi} \cdot (1 - \text{P}_{it}) \quad (8)$$

$$\text{REE PD}_{it} = \text{SFG D}_{it} \cdot \text{REE F}_{i} \cdot (1 - \text{P}_{it}) \quad (9)$$

$D_{i,t48} \cdot IG_{it} \cdot PG_t$ is the demand from 2008 (48 months after starting in the year 2004) in tons and the industrial and population growth in percent at time *t*. Potential demand is reduced by the current price level as expressed by $(1–P_{it})$ to reflect the current price situation of the specific REEs, SFGs or FGs. Equation (8) presents the potential SFG demand where $FG\ D_{it}$ is the absolute demand of the FG in tons and $SFG\ F_i$ reflects the coefficient of the SFGs needed to produce the FGs. For the potential REE demand presented in equation (9), $SFG\ D_{it}$ is the absolute demand of SFGs in tons and $REE\ F_i$ reflects the coefficient of the REEs needed to produce the SFGs.

Regarding the calculation of the industrial growth and the population growth rate, two equations are included in the REE model. The

current industrial growth rate for each FG is determined by a stock that is estimated over time and specific consuming industries that demand the product as:(10)

$$IG_{it} = \sum_{j=1}^{18} \sum_{\tau=t0}^{t} (IGRD_{ij\tau} \cdot DM_{\tau} + IGRE_{ij\tau} \cdot EM_{\tau}) + IG_{i,t0} \quad (10)$$

$IGRD_{ij\tau}$ is the individual growth rate for each FG i at time τ of the specific consumer industry j in the developed market and DM_{τ} is the current allocation factor of the developed market. $IGRE_{ij\tau}$ denotes the growth rate for each FG i at time τ of the specific consumer industry j in the emerging market and EM_{τ} is the current allocation factor of the emerging market. $IG_{i,t0}$ is the initial industrial growth rate of each FG in the year 2004 (in our case 0). The FGs and their consuming industries are listed in the Supplementary Material.

The population growth is based on a non-linear, first-order differential that presents the relative increase of the population in percent and is expressed as:

$$\frac{d}{dx} PG_t = \frac{\Sigma_{\tau=t0}^{t}(IDM_{\tau} + IEM_{\tau}) + Population_{t0}}{Population_{t0}} \quad (11)$$

The counter is a stock that is estimated over time including IDM_{τ} as the net increase of population in developed countries; IEM_{τ} is the net increase of population in emerging countries and $Population_{t0}$ is the initial population in the year 2004 (Population Reference Bureau, 2015; Van Art, 2010; United Nations, 1998). At all three stages, the price level change is calculated based on Sverdrup et al. (2014b), Milton (2008) and Greenman (1994) and is expressed as:

$$\frac{d}{dx} P_{it} = \frac{D_{it} - S_{it}}{S_{it}} \quad (12)$$

Equation (12) presents the relative increase or decrease of the price level in percent by comparing the difference of supply and absolute demand to the supply in one time period. If the absolute demand exceeds supply, the price level increases and vice versa. The price level is captured by the balancing feedback loop B1 as shown in Fig. 1. It is expected that substitutes (R1) and recycled REEs (R2 and R4) will stabilize the price level over time.

Our second module represents R & D and rests upon the chain-linked model by Kline and Rosenberg (1986). They find that innovation processes and R & D activities have to be separated if you wish to use them in a dynamic environment. They introduce the knowledge pool (Technology Potential [TP]) to integrate feedback loops and cover dynamic developments within the chain-linked model. The R & D mechanism is implemented in the REE model at all three stages.

Based on the price level development, political, companies' and economic interests affect the know-how development. Political interest defines a country's motivation to perform basic research. These basic research programs are the foundation on which a specific product, process or recycling technique is developed. The rate of political interest is based on the mining share of the developed country relative to the total global mining amount. The openings of new mines and changes in the mining amount influence the political interest. Equation (13) expresses the political interest:

$$\frac{d}{dx} PI_t = \frac{Developed\ Countries'\ Mining\ Amount_t}{Global\ Mining\ Amount_t} \quad (13)$$

Developed Countries' Mining Amount$_t$ is the mining amount in tons from Australia, Russia and the United States of America and the *Global Mining Amount*$_t$ is the total amount of REEs mined at time *t*.

A company's interests are related to its R & D workforce (RDP). It defines the company's motivation to hire or lay off R & D workforce based on the price level development at each stage for each product. The change in the total R & D workforce is delayed by 6 months as shown in Chamberlain (2015) and Zachert (2004). The R & D workforce is calculated per REE, SFG or FG to reflect varying company behavior and is expressed as:

$$RDP_{it} = Delay\ (6,(1+P_{it})) \quad (14)$$

Finally, the economic interest reflects a company's motivation to invest in R & D and perform a "technology push". It shows the company's motivation for entering the innovation cycle and creating know-how that can be used for reducing risks or developing new market conditions by introducing new products (Kline and Rosenberg, 1986). The economic interest (denoted as EI_{it}) has a static value of 10% and is influenced by the current price level of the REEs, SFGs or FGs.

To meet product characteristics and to enable the development of specific substitutes or recycling techniques, know-how is generated individually for each REE, SFG and FG. It is captured within the balancing feedback loop B2 as shown in Fig. 1. The introduced basic research programs collect fundamental knowledge and trigger innovation. The technology potential is reduced by inventions that iterate a feedback path with back connections directly from market needs and users (Kline and Rosenberg, 1986; Brooks, 1995). The technology potential increases over time per REE, SFG and FG. The current technology potential (denoted as TP_{it}) per REE, SFG and FG is a stock that is estimated as:

$$TP_{it} = \sum_{\tau=t0}^{t} (Delay\ (12, PI_{\tau} \cdot (1+Pi_{\tau})) - I_{i\tau}) + TP_{i,t0} \quad (15)$$

Delay(12, $PI_{\tau} \cdot (1 + P_{i\tau})$*)* is the political interest delayed by 12 months and influenced by the current price level of the REEs, SFGs or FGs, $I_{i\tau}$ is the flow of innovations generated and $TP_{i,t0}$ is the initial technology potential of each REE, SFG and FG in the year 2004 (in our case, there is no TP). The delay value is an average based on Leszczensky et al. (2004).

Innovation takes place based on the existing technology potential, available R & D workforce and economic interest. Innovation is expressed as:

$$I_{it} = TP_{it} \cdot (1+TP_{it} \cdot RDP_{it} \cdot (EI_{it} \cdot (1+P_{it}))) \quad (16)$$

$(EI_{it} \cdot (1 + P_{it}))$ is the economic interest increased by the current price level of *i*.

The know-how rate increases if political interest is high and fundamental research is performed. The know-how illustrates how much information is available for developing substitutes or new recycling techniques. The current know-how rate (denoted as KH_{it}) increases over time per REE, SFG and FG. The know-how rate is a stock that is estimated as:

$$KH_{it} = \sum_{\tau=t0}^{t} I_{i\tau} + KH_{i,t0} \quad (17)$$

$KH_{i,t0}$ is the amount of available know-how at time 0 (in our case no know-how is available) for each REE, SFG and FG.

Our third module represents substitution possibilities at the FG and SFG stages. The substitution amount is based on the potential demand and reduces the absolute demand for SFGs or FGs. Substitutes reduce supply unavailability by using other SFGs that do not request REEs or by selling other FGs that have a different product design to consumers. They have a negative (reducing) effect on the price level. For each SFG and FG, specific substitution rates on the product design are available (Schüler et al., 2011). As shown in Fig. 1, the rebalancing feedback loop R1 illustrates that demand can be reduced by using substitutes. Substitutes are based on the substitution rate and the current potential demand for a product. Substitution rates are based on the findings of Schüler et al. (2011) and Angerer et al. (2009) and given in the Supplementary Material. The substitution rate is multiplied with the current know-how rate of the product to reflect developments. Substitution can be expressed as:

$$Sub_{it} = PD_{it} \cdot Sub\ R_{it} \cdot KH_{it} \quad (18)$$

Sub R_{it} is the specific substitution rate of SFGs or FGs.

Our final module, recycling, is divided into primary and secondary

recycling activities. The rebalancing feedback loop R3 illustrates the primarily recycled amount of REEs, see also Fig. 1. Primary recycling activities take the original REEs back into the production stream without any modification, thus removing them from the SFG production scrap. Secondary recycling focuses on the recovery of REEs out of FG scrap, which requires the disassembly of already manufactured FGs at the end of their life time.

As shown in Fig. 1, the rebalancing feedback loops R2 and R4 illustrate the development of the recycling rate for recycling activities from SFGs and FGs. Each recycling rate develops individually based on the generation of know-how for a specific product. Each SFG and FG has explicit product circumstances, such as for instance complexity of recycling, that make it more or less valuable for recycling.

Recycled REEs out of the SFG scrap are added to the REE supply based on the ratio of the SFG. R5 and R6 show the processing of FGs at the end of their life time and their secondary recycling. Scrap for primary recycling is based on SFG scrap and directly recycled, and then fed back into the REE supply cycle (Glöser et al., 2013). Primary recycling is defined as:

$$\text{REE PR}_{it} = \text{SFG Scrap}_{it} \cdot \text{PC}_i \cdot \text{KH}_{it} \quad (19)$$

SFG Scrap$_{it}$ is the amount that is available for primary recycling in tons and *PC*$_i$*KH*$_{it}$ are the product circumstances that present the degree to which an SFG or FG is recyclable after it has been increased through know-how.

Secondary recycling activities are performed only for 5 out of the 21 FGs as in real life, a recycling infrastructure is present only for 5 FGs. 95% of glass colorized goods, 91% of fluorescent lamps, 33% of all nickel–metal hydride (Ni-MH) batteries, as well as 25% of large motors and wind turbines are collected and available for secondary recycling (Goonan, 2011).

Over time, an FG stock develops as REEs are mined and bound within the global ecosystems. At the end of the life time, some of the produced FGs are available as FG scrap for potential secondary recycling activities. Goonan (2011) states that secondary recycling in low quantities started in 2008. Therefore, we delay the FG scrap by 48 months. The secondary recycled REE amount is added to the REE supply. As 30% of the secondary recycled REEs are of low quality, they have to be smelted again before they are available as REE supply. The secondary recycled amount reduces the FG scrap amount. FGs available as FG scrap reduce the FG stock. The secondary recycling equations can be written as:

$$\text{FG Stock}_{it} = \sum_{\tau=t0}^{t} (\text{FG Pro}_{i\tau} - \text{FG Scrap}_{i\tau}) + \text{FG Stock}_{i,t0} \quad (20)$$

$$\text{REE SR}_{it} = \text{Delay}\,(48, (\text{FG Scrap}_{it} \cdot \text{PC}_i \cdot \text{KH}_{it})) \quad (21)$$

FG Stock$_{it}$ is the current amount of FGs bound in the global ecosystem in tons, *FG Pro*$_{i\tau}$–*FG Scrap*$_{i\tau}$ are those FGs that are available for secondary recycling and *FG Stock*$_{i,t0}$ is the initial amount of available FG stock at time 0 (in our case, no FG stock is available).

Equation (21) presents the secondary recycling amount where *REE SR*$_{it}$ is the absolute amount of REEs secondary-recycled in tons.

To summarize, the balancing loop B1 refers to the dynamics of supply and demand represented in all three stages of the REE model. B2 refers to the dynamics used for generating know-how for each REE, SFG, and FG, all of which are connected to the reinforcing loops R1, R2, and R4. The reinforcing loop R1 refers to the dynamics of generating new substitutes for SFGs and FGs individually. The reinforcing loops R2 and R4 refer to the dynamics of generating new recycling techniques that recollect REEs out of SFG and FG scrap. Finally, the reinforcing loops R3, R5, and R6 refer to the recycling dynamics of reducing SFG and FG scrap, as well as for adjusting the FG stock amount.

2.4. Data collection and compilation

Data are grouped into 11 categories: (1) global mining data, (2) new mines and their planned production amount, (3) REE restrictions, (4) allocation of REEs, (5) design of the 8 REE supply chains and the coefficient at which SFGs and FGs are produced, (6) demand of FGs from 2008, (7) industries calling for products, (8) global population growth, (9) substitution starting rates, (10) available finished goods for secondary recycling and (11) product circumstance factors for each SFG and FG. Data sources include governmental or industrial statistics, like by China's Ministry of Industry and Information Technology (2015), Reichl et al. (2015), Statista (2015), European Commission (2014), United States Geological Survey (2014), United Nations (1998) and books, research reports and academic papers such as Wang et al. (2015), Christmann (2014), Glöser et al. (2013), Du and Graedel (2013), Moss et al. (2013), Goonan (2011), Schüler et al. (2011), van Art (2010), Angerer et al. (2009). Details on data collection and compilation are described in the Supplementary Material.

3. Results

In this section, we test our model against empirical data, describe the applied scenarios to verify our research questions and analyze the sensitivity of the results.

3.1. Model testing

Following the calibration of the REE model, we compare the empirical Neodymium (Nd) price development with the corresponding model output. Simulated values and empirical values between 2004 and 2015 are listed in Table 1. Note that the empirical Nd price development is an annual average price between 2004 and 2008 and a monthly average price starting in 01/2009. We compare the REE Nd price with empirical data as 7 out of 8 REE supply chains use Nd. Due to a lack of market data, other REE prices cannot be compared. The yearly average price between 2004 and 2008 leads to a high error rate that decreases as soon as empirical values are available on a monthly basis. The high error rate is caused by the fact that we discount the FG demand of 2008 to get values for the previous years. No demand data are available for the time between 2004 and 2008. Furthermore, no industry dynamics or their development is included, which might also explain the high error rate. The difference between simulated and real values between 2004 and 2008 vary from −54% to 62% and between 2009 and 2015 from 0.7% to 27%. Besides the empirical fit, structural and behavioral tests following Oliva (2003) and Homer (1983) are used for validating the REE model. We perform partial-model tests by using extreme values to appraise each implemented module individually, which enable us to find out whether the module performs correctly or not. No specific empirical data is available, e.g. there is no know-how rate or substitution rate that can be used for the module validation.

Table 1
Simulated errors of empirical and simulated Neodymium price.

Year	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Empirical values	5.8	6.0	11.2	29.2	32.4	20.1	60.5	316.6	163.1	90.9	91.4	82.3
Simulated values	8.9	7.2	8.1	11.8	12.2	14.6	59.4	314.4	184.9	95.1	92.9	100.8
Error (%)	−53.8%	−20.2%	27.4%	59.7%	62.4%	27.2%	1.8%	0.7%	13.3%	4.6%	1.6%	22.5%

Table 2
Overview of strategies and applied scenarios.

Strategy	Interdependencies	Description of analyzed effects	Scenarios
Substitution	• Development of know-how rate • Development of REE, SFG and FG demand	• Change of substitution rates for REE mining companies, SFG and FG producers • Development of individual REE, SFG, and FG demand	• REE substitution ◦ Normal Rate: 3% starting rate ◦ High Rate: 5% starting rate • SFG substitution ◦ Normal Rate: 3% starting rate ◦ High Rate: 5% starting rate ◦ Earlier start of substitution of 12 periods (T-12) based on normal rate ◦ Impact of R & D on normal rate • FG substitution ◦ Normal Rate: 3% starting rate ◦ High Rate: 5% starting rate ◦ Earlier start of substitution of 12 periods (T-12) based on normal rate ◦ Impact of R & D on normal rate • Combining FG and SFG normal rate substitution scenarios
Recycling	• Development of know-how rate • Development of available SFG and FG scrap	• Impact of primary recycling for SFG producers • Impact of secondary recycling for FG producers • Progress of primary and secondary recycling amount	• Secondary recycling rate ◦ Low Rate: 1% starting rate ◦ Normal Rate: 5% starting rate ◦ High Rate: 10% starting rate • Impact of R & D on normal rate • Sensitivity to REE price with high rate
Opening of new mines	None	• Impact on supply for REE mining companies • Impact on REE price level • Indirect impact on SFG and FG price level	• REE price level impact due to the adding of additional REE supply of 7000 tons in 2012 up to 108,840 tons in 2015
R & D	• Development of individual price level (per REE, SFG and FG) • Development of Political and Economic Interest	• Development of know-how rates for REE mining companies, SFG, and FG companies • Impact on individual primary and secondary recycling rate as well as individual substitution rate for each REE, SFG, and FG	• Active and inactive R & D module to improve FG substitution rate • Active and inactive R & D module to improve SFG substitution rate • Active and inactive R & D module to improve secondary recycling rate • REE price level with and without R & D
• Interplay of Strategies	• Interdependencies between the modeled stages	• Impact on REE price level • Substitution rates for SFG and FG • Primary Recycling rates for SFG • Secondary Recycling rates for FG • Know-how rates for REE, SFG, and FG	• Implementation of Substitution and Recycling strategy with the normal rate setting, the Opening of new mines strategy and an active R & D module to improve FG and SFG substitution rate and secondary recycling rate

3.2. Scenarios

All the different mitigation strategies (see Table 2) are implemented into the REE model one at a time. Substitution spotlights the development of alternative REEs, SFGs, and FGs that reduce the demand for the downstream product. Recycling focuses on the recollection of REEs out of SFG production scrap or end of life for FGs. By opening new mines, REE supply is increased. R & D is utilized to generate know-how and to develop, or further specify, substitutes and recycling techniques. Finally, the interplay of all the four strategies is analyzed by implementing them together, which will help us to understand how the strategies interact and what effects occur.

To analyze the proposed mitigation strategies, different scenarios are implemented and examined. The substitution scenarios cover different starting rates for REE, SFG, and FG substitution. They also show how a change of the delay affects the model and what changes (if any) can be observed from R & D at each stage within the REE model. Recycling scenarios include different starting rates for time, for how R & D affects the recovery amount of secondary recycling and for what mitigation potential recycling has on the REE price. Furthermore, with the increase of supply, we test how the additional REE supply affects the REE price and the downstream stages within the REE model. Substitution and recycling scenarios are implemented and tested with and without R & D to analyze the mitigation potential of R & D and to find out how R & D affects the REE price. Finally, with the interplay of all strategies, we analyze how the REE prices react. Table 2 gives an overview of the scenarios, key variables, and data we used.

3.3. Scenario results

We examine the effects of substitution, recycling, supply expansion and the interplay of all these strategies. A standard way of reading results of system dynamics models is to analyze the behavior of different key variables in a reference condition and compare them with possible scenarios. The reference condition is the normal rate for each strategy. Our alternative scenarios for the relevant strategies are summarized in Table 2.

Figs. 2–6 address the first research question. Effects from substitution, recycling, and a supply expansion differ in their behavior and in their supply unavailability avoidance and price level reduction potential. It is important to note that substitution effects appear in 2004, as opposed to secondary recycling activities that appear in 2008 and supply expansion (2012). Results in Fig. 5 refer to the years between 2008 and 2015, whereas all others were listed from 2004 to 2015, as no

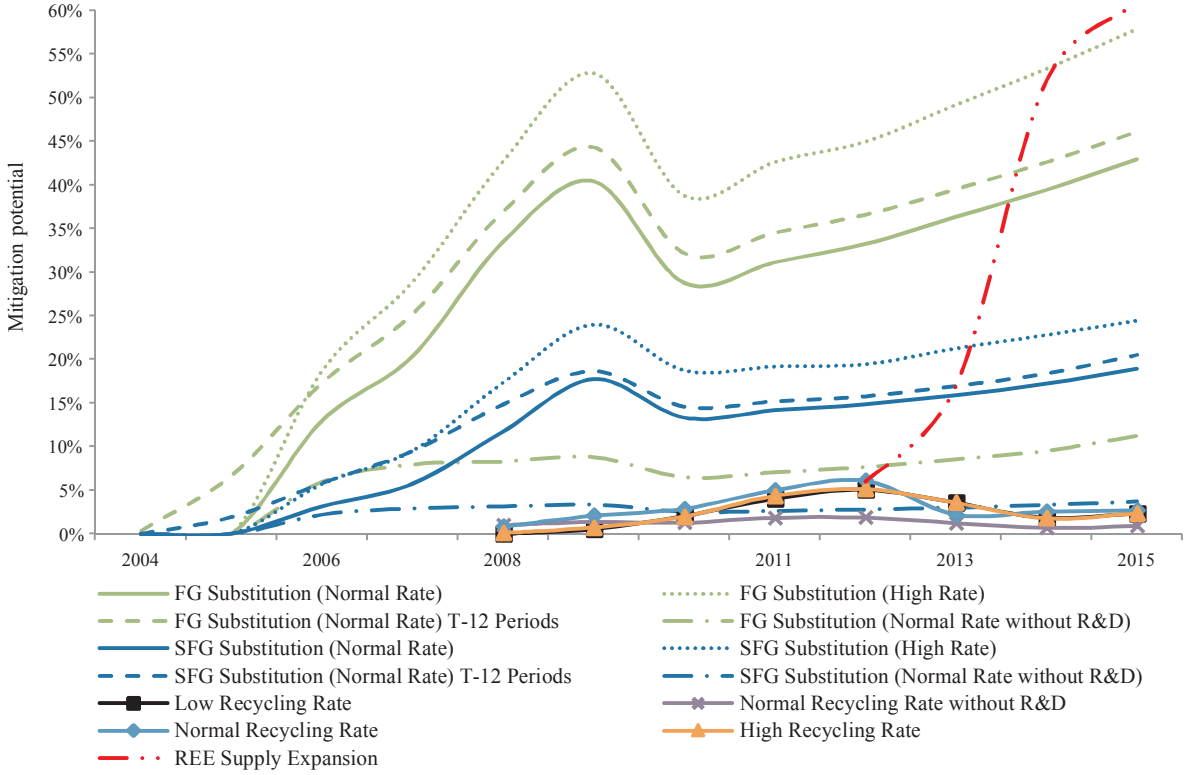


Fig. 2. Weighted mitigation potential for FG substitution, SFG substitution and.

secondary recycling activities were performed before that time.

Fig. 2 shows an index of all weighted SFGs and FGs based on the FG demand in 2008. As illustrated in Fig. 2, the potential of mitigation by avoiding or recovering from supply unavailability between substitution, recycling, and a supply expansion, too, differs. The highest mitigation potential was found for substitution. As illustrated in Figs. 3 and 4, FG producers show a higher substitution rate than SFG companies. Furthermore, FG producers improve their substitution rate faster than SFG producers. Over 5 years, a delta of up to 28% in the mitigation potential for the potential demand can be observed. A scenario analysis for FG and SFG substitution shows that the sensitivity of the applied scenarios, too, differs. A two percent higher starting value leads to a higher mitigation potential of 13% for FG and 6% for SFG producers, respectively.

First, this indicates that FG producers react faster than SFG manufacturers and adjust their product portfolio in less time. A reason for this could be that FG producers have an information advantage regarding the end-consumer demand and their expectation over SFG manufacturers. Second, the development of SFG substitutes up to a point where it achieves a certain mitigation potential takes more time than the development of FG substitutes. Reasons for this could be the REE's specific characteristics. If SFG producers operate in a critical commodity supply chain, they should broaden their product portfolio to reduce dependencies on critical commodities. If this is not possible, the sourcing strategy could be adjusted to reduce country risks if you wish to ensure a stable raw material supply. Note that, if REE substitution is possible, substitution will result in the lowest substitution mitigation rate on the raw material price. However, REE substitution shows a delay of the price effect by 13 months, as illustrated in Fig. 4a. The delay of the REE price increase due to REE substitution indicates that the potential FG demand is progressed through the supply chain stages until the potential demand can be fulfilled. Nevertheless, a substitution strategy for REE mining companies is technically not feasible.

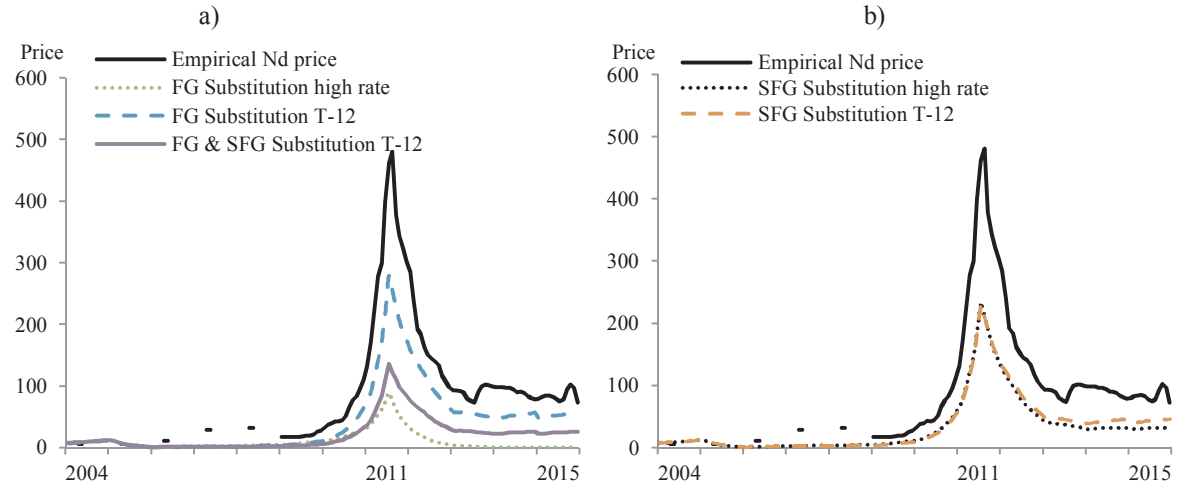


Fig. 3. a) Empirical vs. modeled Nd price with aggregated FG substitution. b) Empirical vs. modeled Nd price with aggregated SFG substitution.

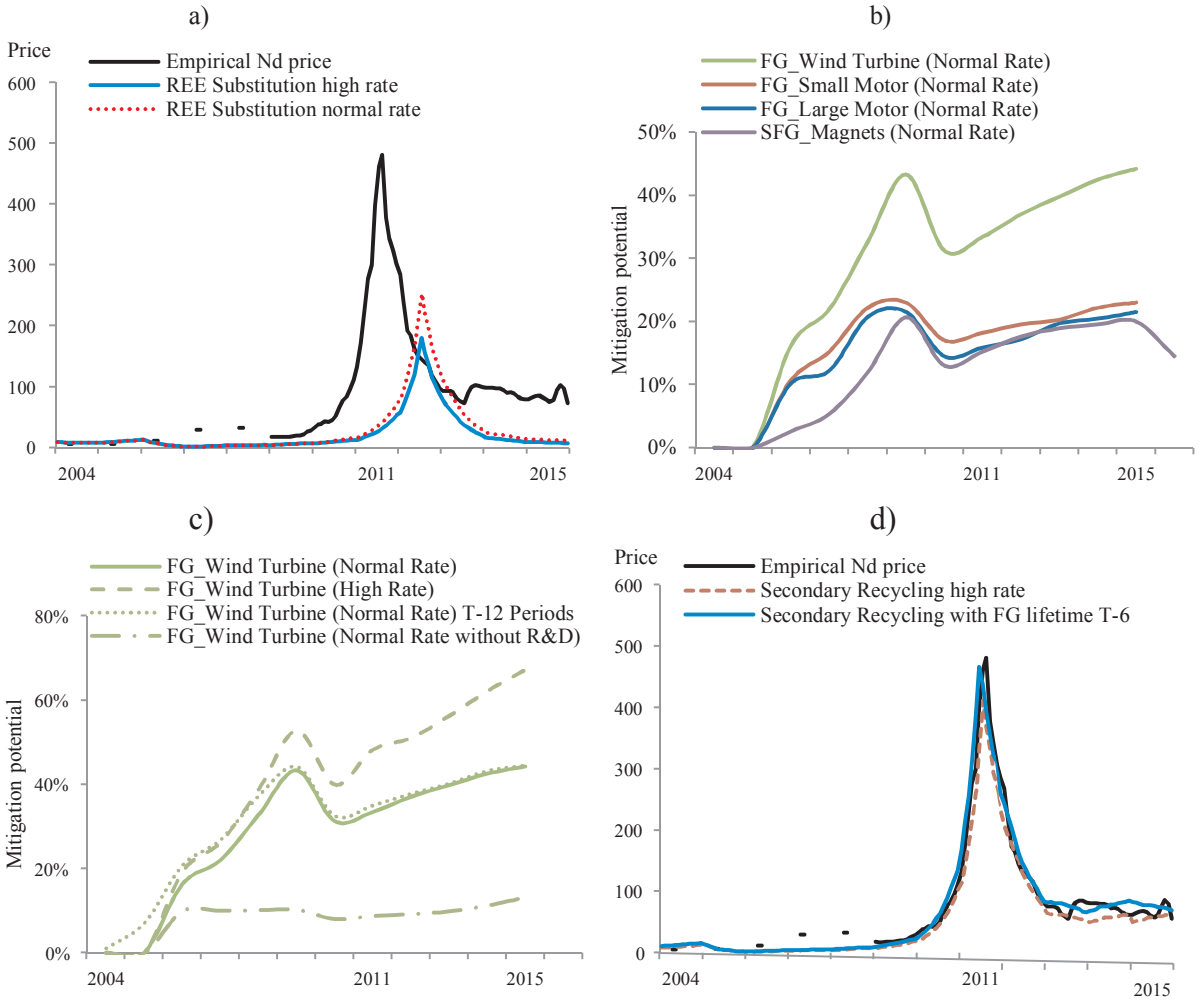


Fig. 4. a) Empirical vs. modeled Nd price with aggregated REE substitution. b) Mitigation potential to avoid supply unavailability from FG and SFG. c) Sensitivity analysis for FG "Wind Turbine" regarding high substitution rate, FG lifetime and R & D. d) Empirical vs. modeled Nd price with aggregated secondary recycling.

REE, SFG and FG substitution results are driven by the development of the substitution rate. A two percent higher starting rate improves the mitigation potential twice as much for FG producers as for the SFG producers. FG producers can take advantage of this leverage effect if they operate in a critical commodity supply chain and apply substitution as their major mitigation strategy.

Secondary recycling has a much lower mitigation potential than FG substitution. As shown in Fig. 4d, the REE price level is only minimally reduced by secondary recycling applied by FG producers. Fig. 5 illustrates that the potential of mitigation through the avoidance of supply unavailability of secondary recycling ranges from 0.1% in 2008 to 4.9% in 2015. A scenario analysis for secondary recycling shows that the sensitivity of the applied scenarios is very small. A ten times higher starting value leads only to a higher secondary recycling mitigation potential of 0.5%, which indicates a very low elasticity for FG producers. However, results from secondary recycling of Ni-MH Batteries demonstrate that a mitigation of the potential FG demand of up to 10.8% can be achieved. This indicates that, if a feasible recycling infrastructure is in place and we have a moderate recycling starting rate of 33%, secondary recycling can be applied by FG producers at a moderate potential for reducing supply

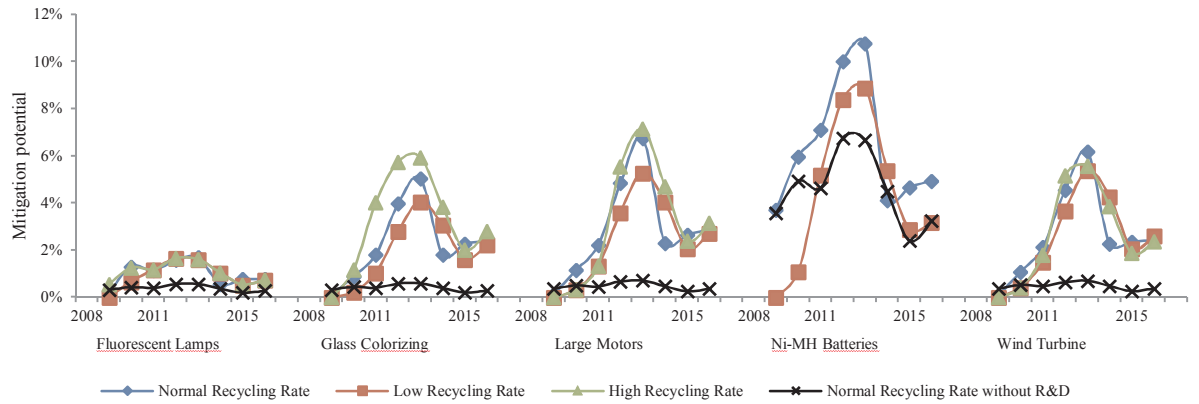


Fig. 5. Mitigation potential to avoid supply unavailability from secondary recycling.

risk. On the other hand, if the secondary recycling rate is low, 10% or less, FG producers should not invest in R & D to increase the recycling rate unless a major improvement is expected as the elasticity between 1% and 10% is of no relevance. The R & D resources should be used for the development of FG substitutes as their mitigation potential illustrates much more promising results.

As illustrated in Fig. 2, supply expansion had a 60% mitigation potential in 2015. However, in our simulation a first new mine was opened in 2012, which operationally impacts SFG and FG manufacturers, causing a delay of 6 years. After 2012, mining capacities of 132,000 tons were progressively added. Due to the opening of new mines in Australia, Brazil, Malaysia, India, South Africa, Tanzania and the USA, the global stakeholder structure changed. China's market share declined from 98.3% in 2009 to 67.4% in 2015, whereas the market share of the other countries, especially Australia and the USA, increased. The high market share of China should have raised a red flag for FG and SFG producers, as well as for governmental institutions. For critical commodities, like REE, it is important to balance raw material supply across the producing countries. A cross-border government-funded mining strategy for critical commodities could have eased the situation from the beginning. For companies that also operate in a critical commodity supply chain, it is highly recommended to analyze which level of supply unavailability risk they have and to proactively start investing into R & D to balance their product portfolio and/or adjust the product design of their goods. Thus, they can exclude these high-risk raw materials as FG and SFG substitution proved to be a feasible mitigation strategy. Furthermore, the governments need to develop a mining strategy for critical commodities that supports the availability of these commodities. In that case, a global evaluation, rather than a country by country comparison, is necessary.

Fig. 6a illustrates how the REE price would have developed if no new supply had been added. Our REE SD model simulates that the raw material price would have peaked at approximately 5000 in the middle of 2012 and declined afterwards until the beginning of 2014, after which time it would again have risen. The REE price increase was driven by the fact that the increasing FG demand could not be met and supply shortages for REE mining companies occurred. Substitution from FG and SFG producers were not able to offset the short-term effect of the rising FG demand. Therefore, the prices for REE increased strongly until the middle of 2012. This also indicates that FG and SFG companies would need time to react to the increasing FG demand and increase the accumulation of know-how, while simultaneously improving the substitution rate to mitigate the potential demand. The decline of the REE price in the second half of 2012 is due to the amount of the demanded FGs having decreased and SFG and FG substitution still being high.

Another important insight is the answer to the question how R & D affects the strategies over time. The behavior of substitution and recycling as described above as opposed to without R & D is shown in Figs. 4 and 5. The Figures illustrate that FG and SFG substitution with R & D has a much higher mitigation potential than without. The price level increase of the FGs causes a strong R & D growth over the price level increase from the SFG. This indicates that FG and SFG producers were not willing to invest into R & D proactively and instead started to increase their R & D investments and activities after export restrictions were introduced.

Answering our third research question, FG producers should apply a substitution strategy to mitigate the risk of REE supply unavailability as FG substitution demonstrated its high mitigation potential and can best influence the REE supply chain if raw material supply is restricted. Furthermore, FG manufacturers should only apply secondary recycling if recycling techniques and a recycling infrastructure are available so that a moderate recycling rate of more than 30% can be achieved. From a substitution perspective, SFG producers have a potentially lower influence when it comes to mitigating potential demand and therefore these companies should broaden their product portfolio in order not to be caught unprepared in cases of raw material unavailability. If this is not possible, they should try to adjust their sourcing strategy so that country risk is reduced. Substitution results for mining companies show the lowest mitigation potential. Due to the specific characteristics of REEs, REE substitution is not realistic in our analysed period and only a theoretical possibility. The accumulation of know-how and consequently the development of the substitution rate for FGs is higher than for SFGs or REEs. Thus it is essential that all companies on all stages of the REE supply chain focus on the development of alternatives, since it was shown in our analysis that R & D significantly improves the mitigation potential gained through substitution and secondary recycling.

Overall, the combination of the mitigation strategies, including R & D, fitted the raw material price best, as is shown in Fig. 6b. The analysis demonstrates that the market combined the different risk mitigation strategies to reduce supply risk caused by export restrictions.

Note that further results of the substitution mitigation potential per supply chain can be found in the Supplementary Material. The results are similar and show the same effects.

3.4. Sensitivity analysis

Using the data of Table 3 for R & D, recycling, and industrial growth, a sensitivity analysis is conducted. Overall, the results show that adjusting the parameters leads only to a minor change in the variables R & D, political interest, secondary recycling and industrial growth.

Sensitivity results for R & D illustrate that changes of the R & D workforce delay have a minor impact on how and when know-how is generated. The chance that R & D will reduce the know-how rate,

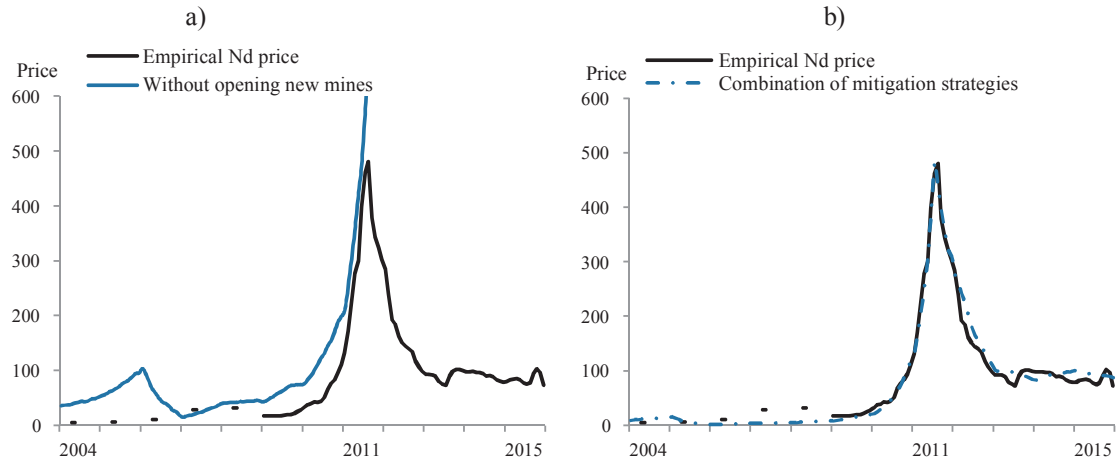


Fig. 6. a) Empirical vs. modeled Nd price without the opening of new mines. b) Empirical vs. modeled Nd price by combining all mitigation strategies with normal rates.

Table 3
Overview of applied sensitivity and their results.

Topic	Sensitivity analysis	Sensitivity results
R & D	• Sensitivity from R & D workforce delay for know-how generation ◦ Normal Case: 6 months ◦ Alternative 1: 3 months ◦ Alternative 2: 9 months • Sensitivity from political interest delay for know-how generation ◦ Normal Case: 12 months ◦ Alternative 1: 8 months ◦ Alternative 2: 18 months	• Average sensitivity for FG and SFG on know-how rate: ◦ vFG: −0.03%; SFG: −0.36% ◦ FG: 1.76%; SFG: 0.67% • Average sensitivity from political interest delay (starting 2012): ◦ -0.005% ◦ 0.011%
Recycling	• Change of HQ REE and LQ REE allocation for secondary recycling ◦ Normal Case: HQ REE 70%; LQ REE 30% ◦ Alternative 1: HQ REE 60%; LQ REE 40% ◦ Alternative 2: HQ REE 50%; LQ REE 50% • Change of FG life time to analyze impact on REE supply	• Average sensitivity on REE supply (starting 2008): ◦ -0.01% ◦ -0.03%
Industry growth	• Sensitivity from demand generation within emerging and developed markets allocation ◦ Normal Case: Emerging markets 42%; Developed markets 58%; Change rate 1.6% p.a. ◦ Alternative 1: Emerging markets 50%; Developed markets 50%; Change rate 1.6% p.a. ◦ Alternative 2: Emerging markets 30%; Developed markets 70%; Change rate 1.6% p.a. ◦ Alternative 3: Emerging markets 42%; Developed markets 58%; Change rate 2.0% p.a.	• Average sensitivity for the industry growth: ◦ 0.34% on Potential FG demand ◦ -0.08% on Potential FG demand ◦ 1.14% on Potential FG demand (as of 2015)

compared to the normal case scenario, varies between 0.03% and 1.76%. For SFGs, it varies between −0.36% and 0.67%. This indicates that the short-term change of the R & D workforce is less important than having a huge number of R & D workers who generate know-how. A change of the political interest delay has almost no influence on how know-how is generated since the political interest is based on the mining share of a country and this share did not change until new mines opened in 2012.

Moreover, results for recycling show that a decrease of 20% for high-quality REEs from FG scrap within the secondary recycling leads to an REE supply that is lower by −0.03% on average. The highest reduction rate is recognized at Praseodymium with −0.06%. On average, the REE price level increased by 0.72%. The highest price level increase is also found for Praseodymium with 3.11%. If the life time of a recyclable FG is reduced by 6 months, the REE supply increases by 2.12% on average. Finally, the sensitivity of the allocation for the industry growth is analyzed. An adjustment of the emerging and developed markets allocation at 50%/50% has only a minor impact on the potential demand of FGs: 0.34% on average. If the rate is adjusted from 1.6% p.a. to 2.0% p.a., the potential demand for FGs increases by 1.14% in 2015. The higher potential demand of FGs has no influence on the REE price level. The same is true for secondary recycling activities and SFG and FG substitution.

4. Discussion and implications

Our scenario results demonstrate the mitigation potential for reducing supply unavailability via substitutes, secondary recycling, and an increase of supply by opening new mines. Furthermore, we show the effects of R & D on the development of substitutes, processes or recycling techniques. In this section, we discuss drawbacks in the context of the applied strategies and uncertainties of the REE market.

4.1. Uncertainties of substitution

In most cases, a simple substitution of assembly groups or modules that contain REEs is impossible (Schüler et al., 2011). Nonetheless, the development of substitutes at all stages was a reaction to the export restrictions introduced by China. Neither SFG nor FG producers had proactively invested in the development of alternative goods (Massari and Ruberti, 2013). The lack of motivation to invest in alternative goods might be justified by the fact that, due to the missing transparency of the REE supply chain, the risk of supply unavailability was not clear to the companies at this point in time. Furthermore, due to the specific characteristics of REEs, a fast and reasonably-priced substitution was not possible. SFGs and FGs require a totally new product design that has to be separately developed, which is time and cost intensive.

Substitutes are nowadays developed for such FGs as new turbines or hybrid vehicles that do not require rare earths. Also, alternatives for SFGs like magnets that do not contain any REEs are developed (United States Department of Energy, 2011). The United States Department of Energy (2011) started a research initiative where alternative products are developed in order to reduce the dependencies on commodities that were identified to be critical in 2011. The initiative was funded with 120 million US-Dollar provided by the US government and runs over 3 years. In line with our findings, substitutes are developed for FGs and SFGs to replace products that include REEs as a way of reducing dependencies from the raw material.

4.2. Uncertainties of recycling

Recycling can contribute towards reducing supply unavailability risk, which affects both primary and secondary recycling activities (Binnemans et al., 2013). However, secondary recycling faces two major challenges. First, for the majority of FGs, no recycling technique exists nor is a feasible recycling technique in place that can recover the bound REEs from existing FG scrap (Guyonnet et al., 2015; Du and Graedel, 2011; Schüler et al., 2011). Second, the FG scrap could also be re-used in other products, for instance recycled lamp phosphors could be used in TVs and for background lightning screens in tablets. There would be no need to recover the REEs out of the FGs as a new market for secondary supply can easily be generated (Balachandran, 2014).

Machacek et al. (2015) performed a value analysis for fluorescent lamps and illustrated that the secondary recycling amount of REEs remains low at 1%. The low recycling amount is in line with our findings as shown in Fig. 5. They state that secondary supply could be a source of supply that makes the industry more independent of China. The improvement of recycling would be in line with many policy goals in countries that want to close the loops of critical materials (Machacek et al., 2015; European Commission, 2014; Moss et al., 2013).

However, the REE market is not capable of improving its efficiency by itself (Ragnarsdottir et al., 2011). Furthermore, a secondary market that covers 1% of the total REE amount has only a limited potential for reducing the dependency from China.

New secondary recycling technologies are developed in academia, but

in most cases they do not yet achieve the break-even point (Schüler et al., 2011). For example, Tan et al. (2017) also focused on fluorescent lamps and developed a new technique to recover REEs. They use mechanical activation in an acid leaching process to recycle Cerium, Europium, Lanthanum, Terbium and Yttrium. Their results show that over 90% of the REEs could be recovered from lamps (Tan et al., 2017). However, no insights regarding a positive business case are available in the study.

The REE prices dropped after 2011 and continue to decline to an even lower level, which makes it difficult for recycling companies that have to invest in the development of new recycling techniques, machinery and the training or hiring of new staff upfront.

Kifle et al. (2013) state that, in the long run, secondary recycling might be very important to ensure REE availability; however, with the aforementioned drawbacks, it is questionable if a feasible recycling is possible. Furthermore, Schüler et al. (2011) estimated that 5–10 years are necessary to set up a REE recycling infrastructure in Europe.

4.3. Uncertainties of supply increase

Because of the REE price drop, there is a high uncertainty about whether or not future REE projects will be developed. Even the current REE price pressure resulted in the bankruptcy of the newly opened REE mine in the USA in 2012 (Brickley, 2016). To start a new mining project, capital and time are required, as it could easily take 10 years before becoming operational (Humphries, 2011). This is in line with our findings, which illustrate that the opening of a new REE mine takes 6 years.

Opening a new mine is one aspect, but processing the REE ore for a product that can be used for SFGs manufacturing requires a high level of know-how about smelting and concentration activities. Currently, only one processing plant for heavy REEs exists; it is located in China (Schüler et al., 2011). Other companies outside of China tried to smelt and concentrate heavy REEs but did not reach the same quality level. Furthermore, China is consolidating its REE industry to meet the REE demand (Machacek et al., 2015). In the future, it is expected that the output of heavy REEs such as Dysprosium, Europium, and Terbium will grow fast with an annual rate of 10% (Christmann, 2014; European Commission, 2014). With the current situation in mind, it is likely that China will continue to dominate the market, especially for heavy REEs.

Furthermore, the balance between economic development and ecological preservation gets more crucial in China. An economic and ecological impact assessment is performed to verify ecological risk and ecological health. For this assessment, dynamic models are used more frequently not only to analyze the current state but also to predict future trends (Liu et al., 2015). As the opening of new mines has a massive ecological impact, it is unlikely that China is going to increase their REE mining. Consequently, mines that have a major negative impact on the environment will be consolidated.

4.4. Uncertainties of companies and political R & D

According to Humphries (2011), investments in R & D are considered to play a critical role in the development of REE substitutes. These findings are in line with our results as, without R & D, the mitigation potential is very low. R & D has been identified to be a key enabler for the development of substitutes, new processes or recycling techniques.

R & D increased the substitution rate for FGs and SFGs by reducing potential demand for all products. However, the development cannot restrict itself to focussing on the raw material composition and its risk profile; it has to fulfill the customers' demand and their needs. Therefore, designing a new product or changing the characteristics of the existing one has to be in line with customer requirements.

The European Union (2014) supports the REE research with 14.5 million Euro through the 7th Framework Programme for Research & Technological Development. The program supports the introduction of raw materials through exploration, processing, recycling and substitution. Basic research is widely applied and fundamental knowledge is developed; however, very few activities are carried out by companies (European Union, 2014). A reason for this is that the costs of R & D and of implementing processes or new recycling techniques are too high. Furthermore, product-specific solutions are necessary for the different kinds of FGs. It is hard to get synergies (European Union, 2014). To reduce the cost pressure, companies could cooperate with each other and split costs, as well as minimize the time to market. In the case of REEs, it would be of interest to investigate how such a coordination would affect the supply chain regarding time, supply unavailability mitigation and the impact on the REE prices. This kind of interaction among companies could give further insights into the dynamic modeling of the R & D strategy (McLellan et al., 2014; Bleiwas and Gambogi, 2013; Milling and Maier, 2011; Stamboulis et al., 2002).

5. Conclusion

Our primary observation after comparing the simulation results for different mitigation strategies is that substitution shows the highest potential for mitigating supply unavailability for SFG and FG producers, followed by secondary recycling for FG producers and the increase of supply by REE mining companies. The chances of avoiding supply unavailability by using a substitution strategy is best for FG companies, whereas SFG companies could also consider primary recycling from SFG scrap as REEs are not yet bound in the product. Our results indicate that SFG producers face more difficulties when trying to develop alternatives and to adjust their product portfolio than FG producers. A reason for this is that there are no alternative raw materials. Secondary recycling demonstrates that, if a suitable infrastructure, suitable processes, and an adequate recycling technique are in place, there is a moderate mitigation potential for FG producers. However, in real life, secondary recycling was only possible for 5 FGs, which indicates that investments into the development of recycling techniques and infrastructure are required urgently. Having to accept a delay of six years before it comes to production means that the increase of supply by REE mining companies has no short-term relevance for SFG and FG producers. The six years delay is mostly caused by preparation activities and necessary environmental and geological tests, both of which are required to get a mining license from the government. From a long-term perspective, it is possible to reduce the high dependency on China in the REE industry as new mines are opened in Australia and the USA. However, the delay also indicates that no cross-border mining strategy for REEs is in place. Our results suggest that establishing a cross-border government-funded mining strategy for critical commodities is an effective strategy, as it was proven that mining companies do not invest proactively in the generation of supply capacities outside China.

A second observation is that the accumulation of know-how is relevant for all FG and SFG producers, as well as for REE mining companies, since it was shown that R & D significantly improves the mitigation potential from substitution and secondary recycling. The results suggest that FG producers should only invest into R & D for secondary recycling if a recycling rate above 30% is feasible. It is crucial for companies and governments to invest proactively into R & D activities, especially where commodity risk is high due to country or geopolitical risk or due to a dependency on a key supplier. The case of the REEs also demonstrates that market power (e.g. export restrictions) is used for influencing a supply chain and for interrupting supply chain robustness.

A third observation is that SFG and FG producers within the REE supply chain chose a combination of different mitigation strategies to react to export restrictions as the interplay of the analyzed strategies offers the best fit for the empirical data. The mitigation potential of the analyzed strategies differs over the three stages. FG producers have a higher effect on the supply chain than SFG manufacturers. As our results indicate, FG producers invest more in the accumulation of know-how than SFG and REE companies.

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Appendix A. Design of the 8 REE supply chains

Within the REE system dynamics model, we have 8 different REE supply chains:
(1) Catalysts & Chemistry, (2) Glass, (3) Metallurgy (excluding battery alloy), (4) Phosphors, (5) Ceramics, (6) Magnets, (7) Batteries and (8) Others as illustrated in Fig. 7. The steps within the three-stage REE model are:

- REE production: mining, concentration, smelting and refining of individual REE oxide. High-quality REE is produced.
- REE processing: Assembling of REE semi-finished goods (SFG). The REE is modified and transformed into a product that is further used in the REE usage step. Primary recycling of semi-finished goods scrap is considered individually for every semi-finished good.
- REE usage: Manufacturing of finished goods (FG) for the specific needs of the requesting industry. Secondary recycling of finished goods scrap at the end of their life time is considered individually for every finished good.
- Industry usage: One or more industry is demanding and consuming finished goods.

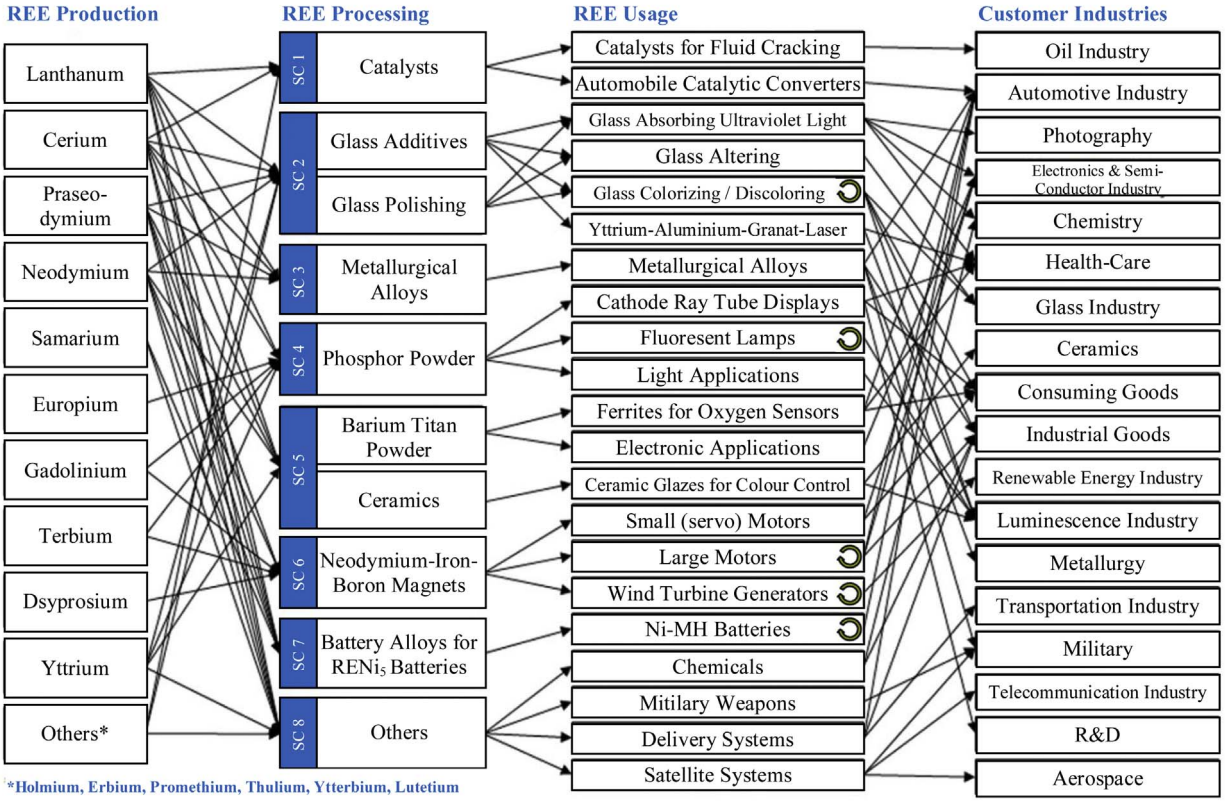


Fig. 7. REE supply chains.

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.resconrec.2017.05.004.

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